

Where and How Fast Is Heat Stress Worsening? A District-Level, Multi-Index Machine-Learning Projection for 64 districts of Bangladesh

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Abstract

Rising heat and humidity increasingly threaten health, labour, and energy systems across Bangladesh, yet operational planning needs district-resolved, uncertainty-aware projections of the specific conditions that drive human heat stress. We present a deterministic, reproducible machine-learning framework that projects seven temperature-correlated heat-stress conditions heat-index comfort, wet-bulb temperature, indoor and outdoor wet-bulb globe temperature, cooling degree-days, ultraviolet exposure, and dew-point moisture stress for all 64 districts of Bangladesh. Four gradient-boosting ensembles are trained on the 2014–2024 monthly record using future-known seasonal features; the observed 1980–2024 Theil–Sen trend is superimposed so that projections evolve realistically, and all thermal indices are recomputed from the projected primary variables to avoid target leakage. Validated against withheld 2025 observations, the projected indices attain median R^2 of 0.95–0.98, and their operational risk classes match observations

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to within one class in 100% of district-months. A Mann–Kendall analysis shows maximum temperature rising in every district (median $+0.56^{\circ}\text{C}$ per decade) while minimum temperature remains flat, indicating a widening diurnal range; the coastal and southwestern zones carry the greatest combined burden, and ultraviolet and oppressive-humidity hazards are effectively nationwide. District choropleths, zonal envelopes, and calibrated uncertainty bands provide a directly actionable heat-stress outlook while transparently bounding the models’ skill relative to climatology. The complete 64-district, seven-condition projection is openly explorable through an accompanying interactive web application.

Keywords: heat stress, wet-bulb temperature, WBGT, seasonal projection, gradient boosting, Mann–Kendall trend, Bangladesh, climate adaptation

1. Introduction

Anthropogenic climate change has fundamentally altered the frequency, intensity, and duration of extreme heat events worldwide. Mean surface air temperatures have risen steadily since the pre-industrial era, and this warming has been accompanied by a disproportionate increase in heatwaves, prolonged humid spells, and compound heat–humidity extremes that the human body increasingly struggles to tolerate [1]. Unlike many climate hazards, extreme heat acts as a silent stressor, producing cumulative physiological strain that is frequently underreported in official mortality and morbidity statistics yet carries substantial public-health, occupational, and economic consequences [2]. Understanding how heat stress manifests across different climatic and geographic settings has therefore become an urgent priority for adaptation planning.

14 Air temperature alone does not represent the heat stress the human
15 body actually experiences and can lead to misleading assessments. Rela-
16 tive humidity, solar radiation, wind, and atmospheric moisture all govern
17 the body’s ability to shed heat through evaporation, radiation, and convec-
18 tion; elevated humidity in particular suppresses the evaporation of sweat
19 the body’s primary cooling mechanism so that core temperature rises even
20 when the air is not exceptionally hot [3]. To capture these combined effects,
21 researchers have developed composite heat-stress indices that now underpin
22 public-health warning systems, occupational heat-safety standards [4], and
23 cooling-energy demand projections [5], and that increasingly recognise the
24 approach of physiological survivability limits under extreme humid heat [6].

25 South Asia is among the regions most acutely exposed to escalating heat
26 stress, owing to its dense population, high monsoon-season humidity, and
27 large share of outdoor and informal-sector labour. Bangladesh occupies a
28 uniquely exposed position within it: a tropical monsoon climate, low-lying
29 deltaic geography, and exceptionally high population density combine with
30 rapid, unplanned urbanization to let heat stress intensify quickly and dis-
31 tribute unevenly across the landscape [7]. Critically, the country is not cli-
32 matically homogeneous its territory spans drought-prone northwestern plains
33 with continental microclimates, a densely built and heat-retentive central
34 urban core exhibiting strong urban heat-island effects, low-lying coastal dis-
35 tricts shaped by maritime humidity, seasonally inundated wetland basins,
36 and elevated hill tracts with distinct topographic regimes. Each zone is
37 expected to experience thermal stress through different pathways and at
38 different intensities, yet most existing heat-risk assessments in Bangladesh
39 continue to treat the country as a single climatic unit or rely on a narrow
40 selection of indices most commonly ambient temperature alone without ac-

41 counting for the multi-dimensional nature of heat exposure [8].

42 A parallel gap exists in temporal scope and forecasting capability. Most
43 prior work on heat stress in Bangladesh is retrospective in orientation or
44 resolved only to divisional or national summaries, and very few studies in-
45 tegrate long-term climatological baselines with near-term machine-learning
46 forecasting to project how these selected temperature correlated conditions
47 will evolve at the district level. This omission matters, because heat-risk
48 management whether through early-warning systems, occupational health
49 policy, or urban cooling infrastructure requires knowing not only where risk
50 is highest today but where and how quickly it is worsening. A recent wet-bulb
51 globe temperature analysis across Bangladesh reported warming of roughly
52 $0.08\text{--}0.5^\circ\text{C}$ per decade, most pronounced in the southeast and northeast [9],
53 underscoring the need for district-resolved, multi-index projection.

54 This study addresses these gaps through the Temperature-Related
55 Conditions (TRC) framework: an integrated set of seven temperature-dependent
56 indicators heat-index comfort, wet-bulb temperature, indoor and outdoor
57 wet-bulb globe temperature, heatstroke risk, cooling degree-days, ultraviolet
58 exposure, and dew-point moisture stress each capturing a distinct physiolog-
59 ical, biophysical, or infrastructural dimension of heat-related risk. Apply-
60 ing this framework to 45 years (1980–2024) of district-level meteorological
61 records across all 64 districts and six climatically distinct zones, we (i) train
62 an ensemble of gradient-boosting models [10] on future-known seasonal fea-
63 tures and re-impose the observed long-term trend so that projections evolve
64 realistically rather than collapsing onto the last training year; (ii) recom-
65 pute every thermal index from the projected primary variables rather than
66 predicting the indices directly, eliminating target leakage and preserving the
67 physical coupling among temperature, humidity, and radiation; (iii) gener-

68 ate 24-month district-level projections (2025–2026) with analytically prop-
69 agated uncertainty bands; and (iv) validate the projections out-of-sample
70 against independent 2025 observations, benchmark their skill transparently
71 against seasonal-naive and climatological baselines, and characterise long-
72 term change with the non-parametric Mann–Kendall test. The best model
73 per variable is selected by a multi-attribute simple-additive-weighting scheme
74 with a generalization-gap penalty [11].

75 Two features distinguish the analysis. First, results are reported at full
76 district resolution through choropleth maps rather than aggregated away,
77 exposing spatial gradients such as the coastal concentration of humid-heat
78 burden that zonal summaries obscure. Second, we are explicit about the lim-
79 its of predictive skill: at monthly resolution the ensembles essentially match
80 climatology, so their value lies in providing spatially complete, uncertainty-
81 aware, and trend-consistent projections of physiologically relevant indices
82 rather than in reducing monthly error. One indicator, heatstroke risk, was
83 found to lack out-of-sample skill and is retained only descriptively. The re-
84 mainder of this paper details the data and methods, presents the validated
85 projections and observed trends, and discusses their implications for heat-
86 adaptation planning across Bangladesh.

87 **2. Literature Review**

88 Human exposure to heat is not adequately captured by dry-bulb air tem-
89 perature alone. A growing body of work therefore evaluates compound ther-
90 mal indices that combine temperature with humidity, radiation, and wind
91 to describe how heat is actually experienced and how it threatens health,
92 labour productivity, and energy demand. This section synthesises recent in-

ternational and Bangladesh-focused literature across the seven temperature-correlated conditions addressed in this study, the machine-learning methods used to forecast them, and the urban and regional context that motivates a district-resolved, multi-index treatment.

2.1. *The physiological basis of thermal-stress indices*

Raymond et al. [6] provided seminal evidence that a wet-bulb temperature (WBT) of 35°C represents the physiological limit for human survival, showing that humid-heat extremes had already more than doubled in frequency between 1979 and 2017 and shifting the field’s focus from ambient temperature toward compound heat–humidity metrics. The inadequacy of dry-bulb temperature as a standalone indicator is well established: high humidity impairs the evaporative efficiency of perspiration, causing perceived loads to exceed measured air temperature [3]. The heat index formalises this through the regression of Rothfusz [12], operationalised by the U.S. National Weather Service. The wet-bulb globe temperature (WBGT), originally developed by Yaglou and Minard [13] for military training and codified in ISO 7243 [4], extends the framework to radiant load, making it the international standard for occupational heat-stress assessment [14]. The empirical WBT approximation of Stull [15], derived by gene-expression programming with a mean absolute error below 0.3°C , provides a computationally efficient estimate applicable to station and reanalysis data alike. Cooling degree-days (CDD) complement these physiological indices by quantifying the cumulative thermal energy above a baseline, a standard proxy for building cooling demand [5, 16]; the UV index [17] captures the solar-radiation dimension of outdoor exposure; and dew-point temperature offers a stable, temperature-independent measure of atmospheric moisture that outperforms relative hu-

midity as a predictor of perceived discomfort [18]. Together these conditions span the physiological, biophysical, and infrastructural dimensions of heat risk that the present study projects jointly, against a backdrop of accelerating global warming documented by the IPCC [19].

2.2. Observed heat-stress trends in Bangladesh

Bangladesh is among the most heat-exposed nations, and a dense body of station-based work documents its warming. Rahman et al. [20] produced the first comprehensive spatiotemporal appraisal of historical and projected heat-index change, finding statistically significant upward trends across most districts, sharpest in the pre-monsoon season. Ekra et al. [21] analysed 60 years (1961–2020) of records from 17 stations using the Discomfort Index, Humidex, and WBT, reporting a spatially uneven rise in heat discomfort that is steepest along the coastal belt and driven primarily by rising air temperature. For WBGT specifically, Kamal et al. [9] provided the anchor analysis, computing outdoor WBGT from 1979–2021 reanalysis with the physically based Liljegren model and finding a national rise of 0.08–0.5 °C per decade, a fivefold expansion of the area affected by high-to-extreme stress, and air temperature as the dominant driver (85.9% relative importance); a companion paper developed simplified regression equations to lower the computational barrier to WBGT adoption [22]. Temperature-extreme indices are likewise intensifying: Ahmed et al. [23] found lengthening warm-spell durations, particularly in coastal regions, and Islam et al. [24] identified warm-index trends dominated by post-2000 increases across 27 stations. Trends in degree-days were examined by Islam et al. [25], who reported significant CDD increases with the southwestern and central zones showing the highest annual means. At the health end, Miah et al. [26] linked ambient temperature and heat in-

145 dex to heatstroke mortality in an ecological time-series analysis, reinforcing
146 the public-health urgency of a multi-index approach.

147 *2.3. Urban heat islands and zone-specific dynamics*

148 Urban heat-island (UHI) intensity amplifies background warming, espe-
149 cially within the densely built central zone examined here. Dewan et al. [27]
150 characterised diurnal and seasonal UHI across five major Bangladeshi cities
151 from satellite land-surface temperature, finding the greatest magnitude in the
152 pre-monsoon period and the largest daytime surface UHI in Dhaka (2.74°C),
153 followed by Chittagong (1.92°C) and Khulna (1.27°C). Faisal et al. [28] re-
154 ported that UHI intensity rose substantially across most districts from 2000
155 to 2020, with Dhaka and Mymensingh steepest, and Park et al. [29] showed
156 that Dhaka’s UHI is strongest in winter (0.95°C) and weakest during the
157 monsoon (0.23°C), with heat waves producing distinctive synoptic patterns
158 that compound local warming. These findings support treating Bangladesh’s
159 urban core as a distinct climatic zone and, more broadly, motivate the six-
160 zone structure adopted in this study.

161 *2.4. Machine learning for meteorological and thermal-index forecasting*

162 Ensemble tree methods have consistently outperformed conventional sta-
163 tistical and numerical approaches for temperature and humidity forecasting
164 in data-rich, strongly seasonal settings. The gradient-boosting family XG-
165 Boost [30], LightGBM [31], and CatBoost [32] recurs as a strong, inter-
166 pretable baseline across environmental prediction problems. In Bangladesh,
167 Mohsin et al. [33] found XGBoost most accurate for seasonal temperature
168 prediction across 29 stations ($R^2 = 0.88$ for maximum and 0.97 for mini-
169 mum temperature) when trained on ENSO predictors; Akter and Rahman

[34] found Random Forest best for humidity forecasting at northern stations over a 40-year record; and Rahman et al. [35] found a stacking ensemble outperformed RNN-LSTM and decision-forest models for average-temperature prediction over a 60-year series. Most directly, Binte Ahmed et al. [36] benchmarked ARIMAX, SARIMAX, Random Forest, XGBoost, and LSTM on 3-hourly Dhaka records (2014–2023) for conditional, scenario-based heat-index projection, with Random Forest achieving the lowest error ($\text{RMSE} = 0.85^\circ\text{C}$, $R^2 = 0.987$) a close methodological precedent for the present work. The value of combining gradient-boosting learners is illustrated by Kim et al. [37], whose stacked XGBoost/LightGBM/CatBoost ensemble reached $R^2 = 0.93$, outperforming any single model. Fourier-based encoding of seasonal cycles as sine and cosine terms has been shown to improve both accuracy and efficiency on meteorological series [38], directly motivating the Fourier feature layer used here. Comparative studies elsewhere reinforce the transferability of these methods [39, 40], while the validation of deep-learning weather models on recent extreme events [41] and analyses of the distinct atmospheric drivers of humidex versus temperature heatwaves [42] situate the forecasting problem in its broader physical context.

2.5. *Per-condition modelling and international context*

Beyond the anchor WBGT and heat-index studies, the individual conditions have distinct literatures. For the heat-index/Humidex family, Sojan and Srinivasan [43] showed that humid-heat stress in India is intensifying disproportionately faster than dry heat; Kushwaha et al. [44] projected air temperature as the dominant driver of future extreme-Humidex days across 15 CMIP6 models; Scoccimarro et al. [45] established that perceived-temperature extremes intensify faster than temperature extremes owing to

196 rising humidity; Saha et al. [46] documented a post-2000 escalation of humid
 197 heatwaves across Northeast India and Bangladesh; and Tabassum et al. [47]
 198 connected a cool-roof intervention to measurable Humidex reductions during
 199 a Dhaka heat wave. Heatstroke-risk modelling has matured chiefly outside
 200 Bangladesh Xu et al. [48] trained interpretable models linking meteorology
 201 to heatstroke incidence in South China with Ehsan et al. [49] providing a
 202 recent Bangladesh-focused machine-learning forecast of extreme-heat health
 203 impact; consistent with this thin evidence base, the present study computes a
 204 heatstroke indicator but, as reported below, excludes it from the validated re-
 205 sults for a lack of out-of-sample skill. For CDD, Bilgili [50] showed that clas-
 206 sical SARIMA models remain competitive with machine learning when the
 207 seasonal structure is strong. UV-index modelling is comparatively underex-
 208 plored in the regional literature: Teggi et al. [51] recently derived the UV in-
 209 dex from reanalysis validated against ground stations, whereas in the present
 210 framework the UV index is instead predicted directly as a modelled vari-
 211 able. Dew-point prediction has a well-developed machine-learning lineage,
 212 progressing from extreme-learning-machine and kernel baselines [52, 53], to
 213 gradient-boosted trees exploiting cross-station information [54], to attention-
 214 based sequence models with explainability [55]. Internationally, the stakes
 215 are large: Parsons et al. [56] estimated humid-heat labour losses equivalent
 216 to roughly 1.7% of 2017 global GDP, with South Asian outdoor workers
 217 among the worst affected, and Ioannou et al. [57] meta-analysed a signif-
 218 icant negative association between WBGT exposure and manual work ca-
 219 pacity. Longer-term projections for Bangladesh under CMIP6 [58] indicate
 220 continued rises in warm-spell duration and tropical nights, providing the
 221 multi-decadal backdrop against which the near-term (2025–2026) forecasts
 222 generated here should be read.

223 2.6. Research gaps addressed by this study

224 Three gaps emerge from this review. First, although individual indices
225 chiefly WBGT and the heat index have been examined for Bangladesh in
226 isolation, no study has simultaneously computed, validated, and forecast
227 seven complementary thermal-stress conditions within a single framework,
228 at district resolution, across climatically distinct zones. Second, existing
229 Bangladeshi forecasting work has relied on conventional statistics, CMIP6
230 downscaling, or single-algorithm machine learning; the multi-algorithm gradient-
231 boosting ensemble with Fourier feature engineering and an explicitly re-
232 imposed observational trend proposed here has not previously been applied
233 to thermal-stress forecasting in this context. Third, the six-zone climato-
234 logical structure of Bangladesh Northwest dry-hot plains, Central urban
235 core, Southwest saline-drought belt, Coastal maritime zone, Haor wetland
236 basins, and Hill Tracts has not previously organised a multi-index heat-risk
237 assessment, leaving intra-national thermal heterogeneity systematically un-
238 dercharacterised. In contrast to studies built on ERA5 reanalysis, the present
239 work is driven by station-derived Visual Crossing records and validated out-
240 of-sample against independent 2025 observations, providing an externally
241 verified, uncertainty-aware, and district-resolved projection across all seven
242 conditions.

243 3. Materials and Methods

244 This study builds a deterministic, reproducible pipeline that (i) learns
245 the monthly climatology of seven primary meteorological variables for each
246 of the 64 districts of Bangladesh from a recent, high-quality window; (ii) su-
247 perimposes the long-term observed climate trend so that projections evolve

248 realistically beyond the training period; (iii) recomputes physiologically rel-
 249 evant thermal indices from the projected primary variables to avoid target
 250 leakage; (iv) quantifies projection uncertainty by first-order error propaga-
 251 tion; and (v) validates the projected 2025 fields against independent obser-
 252 vations. All methodological constants are fixed a priori and are reported
 253 here so that the workflow can be audited and reproduced exactly.

254 3.1. Study area and climatic zonation

255 The analysis covers all 64 administrative districts of Bangladesh. For syn-
 256 thesis and reporting, districts are grouped into six climatic zones: (1) North-
 257 west Extreme Heat Zone (dry hot plains, including the Barind Tract), (2) Cen-
 258 tral Urban Heat Zone (urban heat-island region), (3) Southwest Hot–Dry
 259 Zone (drought- and salinity-affected areas), (4) Coastal Humid Heat Zone
 260 (the coastal belt), (5) Haor/Wetland Humid Zone (the haor basin), and
 261 (6) Hill Tract Zone (elevated terrain). The zonation is author-defined but
 262 grounded in converging, established frameworks: the agro-climatic regionali-
 263 sation of the Bangladesh Meteorological Department and the agro-ecological/physiographic
 264 zonation of Bangladesh [59, 60], together with recognised hydro-ecological
 265 units (Barind Tract, Sundarbans coastal belt, haor basin). Each district
 266 is flagged as a *confirmed* (high-confidence) or *provisional* assignment, and
 267 one representative *exemplar* district per zone is retained for main-text fig-
 268 ures (Figure 1). The complete district-to-zone mapping is provided in the
 269 supplementary material.

270 3.2. Data and preprocessing

271 Daily meteorological records were obtained from the Visual Crossing
 272 Weather API [61] for each district. Seven primary (physically measured)

Climatic zonation of Bangladesh (64 districts, 6 zones)

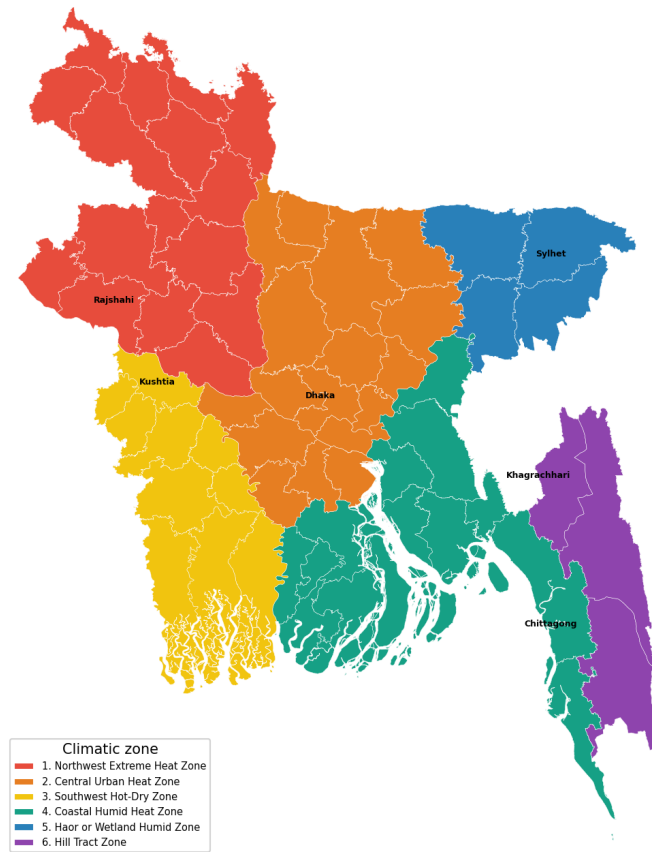


Figure 1: Climatic zonation of Bangladesh: the 64 districts grouped into six zones, with the exemplar district of each zone labelled.

variables are used: mean, maximum, and minimum near-surface air temperature ($^{\circ}\text{C}$); relative humidity (%); dew-point temperature ($^{\circ}\text{C}$); downward solar (shortwave) radiation; and the ultraviolet (UV) index. The historical record spans 1980–2024; an additional file covering 2025 (through 20 November 2025; the final month is partial) is reserved exclusively for out-of-sample validation.

Because humidity is noisy in the earliest decades and solar/UV are unavailable before 2014, the machine-learning window is restricted to 2014–2024, whereas the full 1980–2024 record is used only for the observational trend analysis (Section 3.12). Daily values are aggregated to calendar-month means (month-start resampling), and residual gaps are filled by linear interpolation. All modelling is performed on these monthly series.

3.3. Feature construction

To respect a strict *future-known-features-only* constraint no lags, rolling windows, or autoregressive terms that would be unavailable at projection time each month is described by a deterministic seasonal–secular feature vector. Seasonality is encoded by a truncated Fourier series with period $P = 12$ months and $H = 3$ harmonics,

$$\sin_h(m) = \sin\left(\frac{2\pi hm}{P}\right), \quad \cos_h(m) = \cos\left(\frac{2\pi hm}{P}\right), \quad h = 1, \dots, H, \quad (1)$$

where m is the calendar-month index. The secular component is the calendar year. The full design vector is therefore $\mathbf{x} = [\text{year}, \sin_1, \cos_1, \dots, \sin_3, \cos_3]$ (seven features).

3.4. Predictive models

Four gradient-based tree ensembles are trained independently for every district–variable pair: Random Forest [10], XGBoost [30], LightGBM [31],

297 and CatBoost [32]. Hyperparameters are fixed a priori and applied uniformly
 298 across all districts and variables so that results are mutually comparable
 299 and the pipeline is fully deterministic (global random seed = 42). The fixed
 300 configurations are listed in the supplementary material.

301 3.5. *Trend-aware detrending and projection*

302 Tree ensembles cannot extrapolate beyond the range of their training fea-
 303 tures; if the calendar year is used directly as a predictor, every projected year
 304 collapses onto the last training year, producing a static (trendless) forecast.
 305 To carry the observed climate trend forward while preserving the future-
 306 known constraint, each primary variable is detrended before learning and
 307 re-trended at prediction time. For district–variable series $y(t)$ with continu-
 308 ous time t (in years), a Theil–Sen slope $\hat{\beta}$ is estimated from the annual-mean
 309 series,

$$\hat{\beta} = \text{median}_{i < j} \frac{\bar{y}_j - \bar{y}_i}{t_j - t_i}, \quad (2)$$

310 where $\bar{y}_k$ is the mean of year k . To keep the projected trend consistent with
 311 the manuscript’s observational trend analysis, $\hat{\beta}$ is taken from the robust
 312 1980–2024 record for the five variables that span it (mean/max/min tem-
 313 perature, humidity, dew point); solar radiation and the UV index, which
 314 lack a pre-2014 record, fall back to a Theil–Sen fit over their available 2014–
 315 2024 window. The ensembles are trained on the deseasonalised residual
 316 $r(t) = y(t) - \hat{\beta}t$, so they model only the seasonal signal; the linear trend
 317 is added back at prediction time, $\hat{y}(t) = \hat{r}(t) + \hat{\beta}t$. This design makes each
 318 projected year differ from the previous one by exactly $\hat{\beta}$ and guarantees that
 319 the projected inter-annual tendency equals the reported Mann–Kendall/Sen
 320 slope (Section 3.12). We emphasise that this trend term is included for
 321 physical consistency with the observed record, not to improve short-horizon

accuracy: at monthly resolution the seasonal cycle dominates, so the trend contributes only a small secular adjustment.

The winning model per variable (Section 3.6) is refit on the full 2014–2024 residual and projected forward for a 24-month horizon (January 2025 through December 2026). Projected primary variables are clipped to physically plausible bounds before any downstream computation to prevent non-physical extrapolations from propagating into the indices.

3.6. Model evaluation and selection

Reported test metrics use a strictly chronological hold-out: models are trained on 2014–2022 and tested on the withheld 2023–2024 period. Generalisation is additionally assessed by five-fold `TimeSeriesSplit` cross-validation on the residual series. For each candidate model we record the coefficient of determination R^2 and root-mean-square error (RMSE) on the chronological test set, the cross-validated R^2 ($CV\text{-}R^2$), a generalisation gap $\text{Gap} = |R^2 - CV\text{-}R^2|$, and an operational tolerance accuracy A_τ defined as the fraction of test months whose absolute error does not exceed a variable-specific tolerance τ ,

$$A_\tau = \frac{1}{N} \sum_{n=1}^N \mathbf{1}(|y_n - \hat{y}_n| \leq \tau). \quad (3)$$

The winning model is chosen by Simple Additive Weighting (SAW), a multi-attribute decision-making scheme [11], with a generalisation-gap penalty. Each criterion c is min-max normalised across the four candidates directly for benefit criteria and by complement for cost criteria,

$$\tilde{s}_c = \begin{cases} \frac{s_c - \min_c}{\max_c - \min_c}, & c \text{ maximised,} \\ \frac{\max_c - s_c}{\max_c - \min_c}, & c \text{ minimised,} \end{cases} \quad (4)$$

and the composite score is the weighted sum $S = \sum_c w_c \tilde{s}_c$, with weights $w_{R^2} = 0.25$, $w_{CV-R^2} = 0.25$, $w_{RMSE} = 0.20$, $w_{Gap} = 0.20$, and $w_{A_\tau} = 0.10$ (RMSE and Gap treated as cost criteria). The highest-scoring model wins; models with Gap > 0.10 are additionally flagged as overfitting risks.

3.7. Thermal index computation

To avoid target leakage, thermal-stress indices are never predicted directly; they are recomputed analytically from the *projected* primary variables. The heat index (HI) uses the U.S. National Weather Service Rothfus regression [12]. With temperature T_F and relative humidity R (in °F and %),

$$\begin{aligned} \text{HI} = & -42.379 + 2.04901523 T_F + 10.14333127 R - 0.22475541 T_F R \\ & - 6.83783 \times 10^{-3} T_F^2 - 5.481717 \times 10^{-2} R^2 + 1.22874 \times 10^{-3} T_F^2 R \\ & + 8.5282 \times 10^{-4} T_F R^2 - 1.99 \times 10^{-6} T_F^2 R^2, \end{aligned} \quad (5)$$

with the standard low-humidity and high-humidity adjustments applied within their validity ranges and a simplified formula substituted below 80°F; the result is returned in °C. Wet-bulb temperature T_w follows the empirical formulation of Stull [15],

$$\begin{aligned} T_w = & T \arctan[0.151977 (R + 8.313659)^{1/2}] + \arctan(T + R) \\ & - \arctan(R - 1.676331) + 0.00391838 R^{3/2} \arctan(0.023101 R) - 4.686035, \end{aligned} \quad (6)$$

with T in °C and R in %. Wet-bulb globe temperature is computed in a shade variant (no direct-beam term) and a sun variant that adds a solar-

359 radiation contribution S ,

$$\text{WBGT}_{\text{shade}} = 0.7 T_w + 0.3 T, \quad (7)$$

$$\text{WBGT}_{\text{sun}} = 0.7 T_w + 0.2 (T + 0.025 S) + 0.1 T. \quad (8)$$

360 Cooling degree-days are expressed as a monthly-mean daily rate about an
361 18°C base,

$$\text{CDD} = \max\left(\frac{T_{\text{max}} + T_{\text{min}}}{2} - 18, 0\right). \quad (9)$$

362 A composite heatstroke-hazard score (a sigmoid of the heat index mod-
363 ulated by a wet-bulb multiplier) was also implemented and recalibrated for
364 monthly-mean inputs. However, in external validation it produced a nega-
365 tive out-of-sample R^2 (i.e. worse than a constant predictor), so it is *excluded*
366 *from all predictive results* and retained only as a descriptive projected field.
367 This exclusion is stated explicitly to avoid over-interpretation of an index
368 that does not demonstrate skill.

369 3.8. Operational heat-stress conditions and risk categorisation

370 The projected fields are organised into seven temperature-correlated heat-
371 stress conditions of operational relevance: (1) heat-index comfort (perceived
372 temperature under humidity); (2) wet-bulb temperature (the thermody-
373 namic limit of evaporative cooling); (3) wet-bulb globe temperature in in-
374 door (shade) and outdoor (sun) forms (the occupational heat-exposure stan-
375 dard); (4) heatstroke risk; (5) cooling degree-days (a building-cooling en-
376 ergy proxy); (6) ultraviolet exposure risk; and (7) dew-point moisture stress.
377 Conditions (6) and (7) are obtained directly from the projected UV-index
378 and dew-point fields, which are among the primary modelled variables (Sec-
379 tion 3.2); condition (4) is excluded from predictive results as noted above.

380 To move from continuous values to operational meaning, four of the
 381 conditions are additionally mapped to established risk-category systems:
 382 the U.S. National Weather Service heat-index comfort classes (Comfortable,
 383 Caution, Extreme Caution, Danger, Extreme Danger); WBGT occupational-
 384 exposure classes (Low, Moderate, High, Extreme); the World Health Orga-
 385 nization UV-index classes (Low, Moderate, High, Very High, Extreme) [17];
 386 and dew-point human-comfort classes (Dry through Extremely Oppressive).
 387 Wet-bulb temperature (2) and cooling degree-days (5) are reported as con-
 388 tinuous physical quantities. The category boundaries are conventionally de-
 389 fined for daily peak values; because the present pipeline operates on monthly
 390 means, the extreme classes are seldom reached and the categories should be
 391 read as describing *typical monthly* conditions rather than daily peaks. Cat-
 392 egorisation skill is assessed against the observed 2025 fields by exact-match
 393 agreement and by agreement to within one adjacent class.

394 3.9. Uncertainty quantification

395 Projection uncertainty for the temperature–humidity indices is obtained
 396 by first-order (linearised) error propagation from the primary-variable un-
 397 certainties. For an index $I(x_1, \dots, x_k)$ with input standard deviations σ_{x_j} ,

$$\sigma_I = \left[\sum_j \left(\frac{\partial I}{\partial x_j} \right)^2 \sigma_{x_j}^2 \right]^{1/2}, \quad (10)$$

398 where the input σ_{x_j} are taken from each winning model’s in-sample RMSE
 399 and representative sensitivity coefficients are used for the heat index, wet-
 400 bulb temperature, and WBGT. The propagated σ_I define the 68% ($\pm\sigma_I$)
 401 and 95% ($\pm 1.96\sigma_I$) bands reported for the projected indices.

402 *3.10. Baseline skill benchmarks*

403 Model skill is benchmarked against two references evaluated on the same
 404 chronological test set: a seasonal-naive predictor (each test month set to the
 405 same calendar month one year earlier) and a monthly-climatology predic-
 406 tor (the training-period mean for each calendar month). Skill relative to a
 407 reference is

$$SS = 1 - \frac{RMSE_{\text{model}}}{RMSE_{\text{ref}}}, \quad (11)$$

408 so that positive values indicate improvement over the reference.

409 *3.11. External validation against observed 2025*

410 The 24-month projection includes calendar year 2025, for which inde-
 411 pendent observations were withheld from all training. Projected monthly
 412 primary variables and recomputed indices are compared with the observed
 413 2025 monthly series using R^2 , RMSE, mean absolute error, and the tolerance-
 414 accuracy bands of Section 3.6. The final month (November 2025) is partial
 415 in the source data and is treated accordingly. This constitutes a genuine,
 416 fully out-of-sample test of the projection.

417 *3.12. Observational trend analysis*

418 Long-term trends are characterised on the full 1980–2024 annual-mean
 419 series using the non-parametric Mann–Kendall test with Theil–Sen slope
 420 estimation, computed independently of the predictive models (and therefore
 421 free of target leakage). The Mann–Kendall statistic is

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^n \text{sgn}(x_j - x_k), \quad (12)$$

422 with tie-corrected variance

$$\text{Var}(S) = \frac{n(n-1)(2n+5) - \sum_i t_i(t_i-1)(2t_i+5)}{18}, \quad (13)$$

where t_i is the number of ties of extent i . The standardised statistic

$$Z = \begin{cases} (S - 1)/\sqrt{\text{Var}(S)}, & S > 0, \\ 0, & S = 0, \\ (S + 1)/\sqrt{\text{Var}(S)}, & S < 0, \end{cases} \quad (14)$$

is referred to the standard normal distribution for a two-sided p -value, and Kendall's $\tau = 2S/[n(n - 1)]$ measures rank correlation with time. The Theil–Sen slope (Eq. 2) provides the trend magnitude, reported per decade. Trends with $p < 0.05$ are deemed significant. This analysis is performed for mean, maximum, and minimum temperature, humidity, dew point, and the temperature–humidity indices (heat index, wet-bulb temperature, shade WBGT) that are computable across the full record.

A subset of districts encode pre-~2000 missing days as zero in the source record; left uncorrected, the resulting $0 \rightarrow \sim 30^\circ\text{C}$ step fabricates physically impossible trends (an artefactual $+7.8^\circ\text{C}$ per decade in one district). Before any trend is fit, daily values below physically plausible floors (5°C for temperatures, with lower floors for minimum temperature and dew point, and 5% for humidity) are therefore treated as missing, and a year must retain at least 60 valid days to enter the annual-mean series. The same masked slopes are used when the trend is re-added to the projection (Section 3.5). This affects only the observational trend and long-run slope; the machine-learning models are unaffected because they train exclusively on the complete 2014–2024 window.

3.13. Zonal synthesis

District-level projections are summarised by zone as monthly and annual envelopes (the minimum, mean, and maximum across member districts) for

the key indices, providing compact, journal-page-efficient views of the projected heat-stress distribution in place of 64 individual district panels. Full district-level detail is additionally retained through choropleth maps rendered directly from the CC BY 3.0 IGO geoBoundaries Bangladesh ADM2 administrative boundaries [62]; districts are matched to the boundary geometry by name, and per-district values are mapped for the observed trends, the projected annual means of all seven conditions, and the number of months per year each district spends in each condition’s hazard class.

3.14. Implementation and reproducibility

The pipeline is implemented in Python 3.14 using scikit-learn 1.8 [63], LightGBM 4.6, XGBoost 3.3, CatBoost 1.2.10, together with NumPy, pandas, and Matplotlib. All stochastic components use a fixed seed, and a run manifest records package versions, configuration constants, and per-run timings so that every reported artefact is traceable to a specific configuration. The complete source code, the reviewer workbook, and an interactive web application that exposes every district and condition are released with the study (see Data availability).

4. Results

The full pipeline was executed for all 64 districts, producing 448 district-variable models, a 24-month projection (January 2025–December 2026) per district, an out-of-sample validation against observed 2025, and the 1980–2024 observational trend analysis. Unless stated otherwise, statistics are reported as the median across the 64 districts. Because the complete per-district projections, validations, and risk categories for all seven conditions cannot be reproduced in print, they are additionally provided through an

470 interactive web dashboard¹ through which any district and condition can be
471 explored directly.

472 *4.1. Predictive accuracy and model selection*

473 On the chronological 2023–2024 hold-out, the ensembles reproduced the
474 primary variables accurately, with median test R^2 of 0.970 for minimum
475 temperature, 0.961 for mean temperature, 0.962 for dew point, and 0.886 for
476 maximum temperature; humidity (0.632), solar radiation (0.691), and the
477 UV index (0.664) were intrinsically harder and scored lower. Under the SAW
478 criterion, LightGBM was selected most frequently (284 of 448 fits), followed
479 by Random Forest (111) and CatBoost (53); XGBoost, although included in
480 the candidate pool, was never top-ranked. Model preference was variable-
481 dependent (Table 1): LightGBM dominated temperature (57/64), maximum
482 temperature (63/64), and dew point (48/64), whereas Random Forest was
483 preferred for the noisier humidity field (55/64) and was competitive for solar
484 radiation and UV.

485 *4.2. External validation against observed 2025*

486 The projected 2025 fields, produced without any exposure to 2025 obser-
487 vations, validated strongly (Table 2). Median R^2 reached 0.982 for minimum
488 temperature, 0.980 for dew point, and 0.950 for mean temperature (RMSE
489 0.72 °C). Crucially, the thermal-stress indices recomputed from the projected
490 primary variables rather than predicted directly also validated well: wet-
491 bulb temperature and shade WBGT both at $R^2 = 0.969$ (RMSE ≤ 0.63 °C),
492 sun WBGT at 0.966, the heat index at 0.951 (RMSE 1.36 °C), and cooling
493 degree-days at 0.951. Maximum temperature (0.845), humidity (0.825), UV

¹<https://tcc-64-districts-mohammad-khalid.streamlit.app/>

Table 1: Best-model selection frequency across the 64 districts, by variable (SAW criterion). XGBoost was never selected and is omitted.

Variable	LightGBM	Random Forest	CatBoost
Mean temperature	57	0	7
Maximum temperature	63	0	1
Minimum temperature	44	2	18
Relative humidity	4	55	5
Dew point	48	0	16
Solar radiation	29	29	6
UV index	39	25	0

(0.805), and solar radiation (0.766) were the weakest, consistent with their lower learnability. For the six exemplar districts, heat-index validation R^2 ranged from 0.873 (Sylhet) and 0.891 (Chittagong) to 0.948–0.954 (Kushtia, Khagrachhari, Rajshahi, Dhaka), with RMSE between 1.1 and 1.9 °C.

4.3. Skill relative to baselines

Against the seasonal-naive reference the ensembles were skilful in 81.5% of district–variable cases, with the largest median gains for minimum temperature (+0.24) and dew point (+0.21). Relative to monthly climatology, however, the models were essentially at parity: the median skill score was near zero or slightly negative for every variable (e.g. −0.003 for mean temperature, −0.030 for maximum temperature, −0.059 for minimum temperature), and only 31.9% of cases exceeded climatology. This is the expected behaviour at monthly resolution: the ensembles reconstruct the strong tropical seasonal cycle that monthly climatology already encodes, so they add little in raw error terms. Their contribution is therefore not a reduction in

Table 2: External validation of the projected 2025 fields against observed 2025, summarised as the median across 64 districts. Errors are in each variable's native unit ($^{\circ}\text{C}$ for temperatures and thermal indices; % for humidity; W m^{-2} for solar radiation; index units for UV; $^{\circ}\text{C d d}^{-1}$ for CDD).

Variable	n	Median R^2	Median RMSE	Median MAE
Mean temperature	64	0.950	0.720	0.615
Maximum temperature	64	0.845	0.940	0.788
Minimum temperature	64	0.982	0.565	0.449
Relative humidity	64	0.825	2.334	1.996
Dew point	64	0.980	0.601	0.469
Solar radiation	64	0.766	14.352	11.704
UV index	64	0.805	0.375	0.302
Heat index	64	0.951	1.357	1.142
Wet-bulb temperature	64	0.969	0.624	0.509
WBGT (shade)	64	0.969	0.595	0.506
WBGT (sun)	64	0.966	0.644	0.543
Cooling degree-days	64	0.951	0.696	0.577

509 monthly RMSE but a spatially complete, uncertainty-quantified, and trend-
510 consistent projection that climatology cannot provide.

511 *4.4. Observed long-term trends, 1980–2024*

512 The Mann–Kendall/Sen analysis revealed a coherent warming signal (Ta-
513 ble 3, Figure 2). Maximum temperature increased significantly in all 64 dis-
514 tricts (median $+0.56^{\circ}\text{C}$ per decade), and the heat index, wet-bulb temper-
515 ature, and shade WBGT increased in 58, 57, and 58 of 64 districts (median
516 $+0.37$, $+0.17$, and $+0.17^{\circ}\text{C}$ per decade, respectively). Mean temperature
517 rose in 50 districts and dew point in 46. Minimum temperature was markedly
518 more heterogeneous (16 increasing, 13 decreasing, 35 without significant
519 trend; median $+0.02^{\circ}\text{C}$ per decade), and humidity was mixed (24 increas-
520 ing, 23 decreasing). The sharp contrast between near-universal maximum-
521 temperature warming and the essentially flat minimum-temperature signal
522 indicates a widening diurnal temperature range in much of the country; a
523 small number of districts (e.g. Sylhet) therefore show a slight decrease in
524 *mean* temperature despite rising maxima. Because the projection re-imposes
525 exactly these Sen slopes, the projected inter-annual tendency reproduces this
526 observed signal by construction.

527 *4.5. Projected 2025–2026 outlook and zonal synthesis*

528 Consistent with the modest observed slopes, the projected year-over-year
529 change was small: the heat index rose by a median of $+0.038^{\circ}\text{C}$ per year
530 across the 64 districts (range -0.034 to $+0.096^{\circ}\text{C}$ per year), the negative
531 values corresponding to the districts whose 1980–2024 mean-temperature
532 trend is itself slightly negative. The zonal synthesis (Table 4) places the
533 highest projected annual-mean heat index in the Southwest Hot–Dry Zone

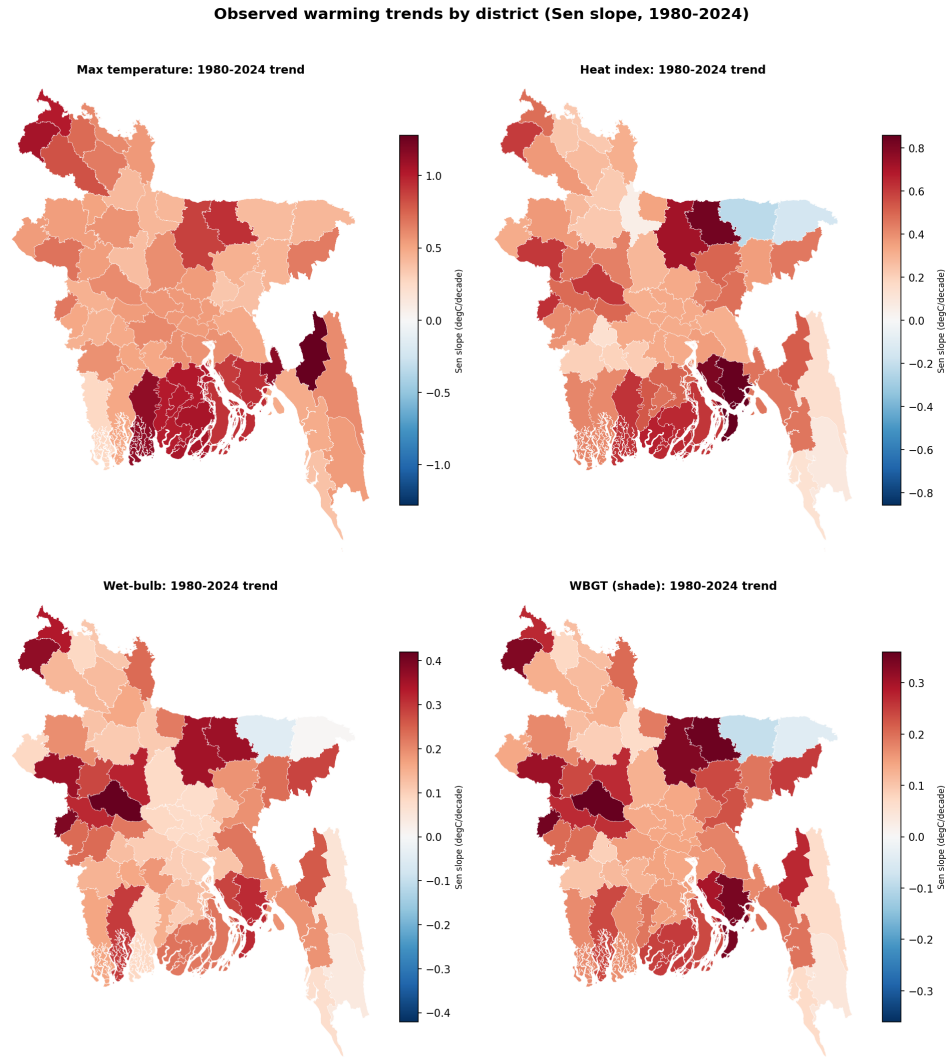


Figure 2: Observed 1980–2024 warming by district (Theil–Sen slope, °C per decade) for maximum temperature, heat index, wet-bulb temperature, and shade WBGT. Red indicates warming, blue cooling. Values are computed after masking missing-as-zero early-record days (Section 3.12).

Table 3: Observed 1980–2024 trends (Mann–Kendall test with Theil–Sen slope) by variable: number of districts increasing, decreasing, or without a significant trend ($p < 0.05$), and the median Sen slope per decade ($^{\circ}\text{C}$ per decade, except humidity in % per decade).

Variable	Increasing	Decreasing	No trend	Median slope/decade
Maximum temperature	64	0	0	+0.558
Mean temperature	50	2	12	+0.194
Minimum temperature	16	13	35	+0.021
Dew point	46	0	18	+0.197
Relative humidity	24	23	17	+0.177
Heat index	58	1	5	+0.373
Wet-bulb temperature	57	0	7	+0.165
WBGT (shade)	58	0	6	+0.172

534 (31.0 $^{\circ}\text{C}$) and the Coastal Humid Heat Zone (30.6 $^{\circ}\text{C}$), and the lowest in
 535 the Haor/Wetland Zone (28.0 $^{\circ}\text{C}$); sun WBGT peaked in the coastal belt
 536 (25.5 $^{\circ}\text{C}$). The two continuous-burden conditions reinforce this ordering:
 537 projected cooling degree-days a proxy for building cooling-energy demand
 538 were highest in the Southwest Hot–Dry and Coastal Humid Heat zones
 539 (9.1 and 9.0 $^{\circ}\text{Cdd}^{-1}$) and lowest in the Haor/Wetland Zone (7.0), while
 540 projected wet-bulb temperature, the thermodynamic ceiling on evaporative
 541 cooling, followed the same spatial pattern, peaking in the Coastal (23.7 $^{\circ}\text{C}$)
 542 and Southwest (23.6 $^{\circ}\text{C}$) zones and again reaching its minimum in the Haor
 543 basin (22.6 $^{\circ}\text{C}$). The coastal belt therefore emerges as the zone of great-
 544 est combined perceived-heat, humid-heat, and cooling-energy burden. Fig-
 545 ure 6 maps all six continuous conditions at district resolution, confirming the
 546 coastal-south concentration of heat, humidity, and cooling demand. Figure 3

547 illustrates the full record-validation-projection narrative for the Dhaka ex-
548 emplar, and Figure 4 summarises the projected monthly climatology of all
549 six key conditions across the six zones, exposing their distinct seasonal sig-
550 natures. Figure 5 shows the district spread within the hottest (Southwest)
551 zone.

Table 4: Projected annual-mean thermal indices by climatic zone (zone mean, with the across-district minimum and maximum in parentheses). Values in °C except CDD (°Cdd⁻¹).

Zone	Heat index	WBGT (sun)	Wet-bulb	CDD
1. Northwest Extreme Heat	29.6 (28.4–30.7)	24.7	22.9	8.3
2. Central Urban Heat	29.6 (29.3–30.9)	24.8	22.8	8.8
3. Southwest Hot–Dry	31.0 (30.6–31.9)	25.5	23.6	9.1
4. Coastal Humid Heat	30.6 (29.5–31.5)	25.5	23.7	9.0
5. Haor/Wetland Humid	28.0 (26.0–29.5)	24.3	22.6	7.0
6. Hill Tract	29.9 (29.3–30.3)	25.4	23.5	8.6

552 4.6. Operational risk categories

553 Mapping the projected fields onto their risk-category systems yields a
554 policy-facing summary of the seven conditions (Table 5). Under monthly-
555 mean heat index, projected months split between Comfortable (37.8%) and
556 Extreme Caution (51.8%), with the intermediate Caution class at 10.2% and
557 the Danger class essentially absent (0.1%) the latter a direct consequence of
558 categorising monthly means rather than daily peaks. Outdoor WBGT was
559 predominantly Low (59.0%) to Moderate (37.0%), reaching the High class
560 in 3.9% of months, while indoor (shade) WBGT rarely left the Low class
561 (79.8%). The UV signal was the most consistently hazardous condition:

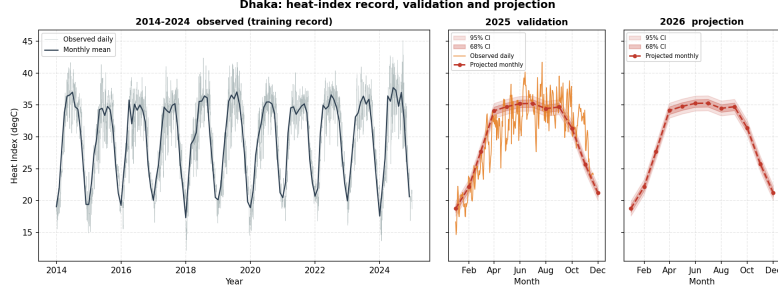


Figure 3: Heat-index record, validation, and projection for the Dhaka exemplar: (left) observed daily heat index 2014–2024 with the monthly mean; (centre) the 2025 validation year, observed daily versus the monthly projection with 68% and 95% confidence bands; (right) the 2026 monthly projection with confidence cloud.

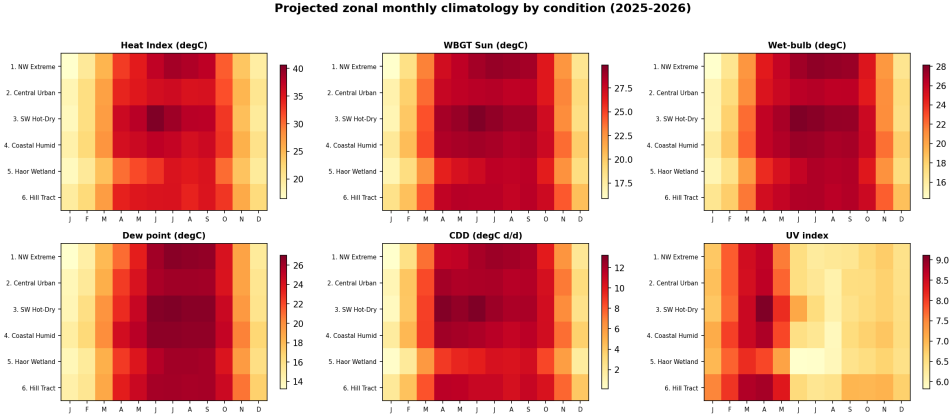


Figure 4: Projected zonal monthly climatology for six heat-stress conditions (heat index, sun WBGT, wet-bulb temperature, dew point, cooling degree-days, and UV index). The temperature-driven conditions peak in the pre-monsoon to monsoon window, whereas the UV index peaks in the dry pre-monsoon (March–April) and falls during the cloudy monsoon a distinct, radiation-driven seasonality.

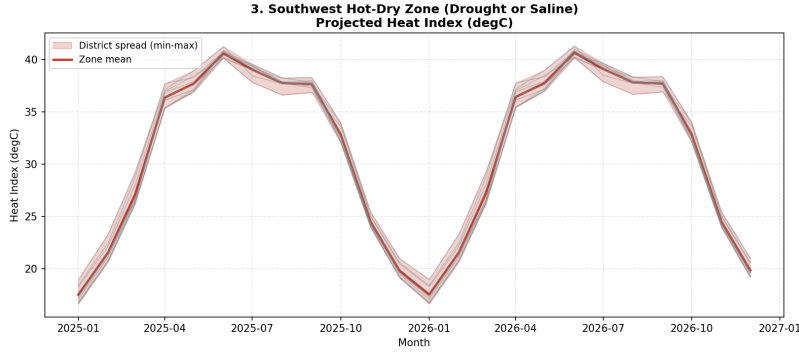


Figure 5: Projected heat index for the Southwest Hot-Dry Zone: the zone mean and the min-max envelope across member districts, illustrating within-zone spread.

562 75.5% of projected months fall in the WHO High class and a further 20.4% in
563 Very High. Dew-point moisture stress was also severe, with 45.5% of months
564 in the Extremely Oppressive class. Across zones, the Southwest Hot-Dry
565 Zone carried the largest Extreme-Caution share for heat index (57.5%) and
566 the only appearance of the Danger class (0.8%), whereas the Haor/Wetland
567 Zone was the mildest (35.4% Extreme Caution); Figure 7 shows the full zonal
568 composition and Figure 8 maps, for each district, the number of months per
569 year spent at or above each condition's hazard threshold. The months-per-
570 year view resolves the strong spatial and seasonal gradients that a dominant-
571 class map obscures: the coastal south endures dew-point levels in the Ex-
572 tremely Oppressive class for roughly seven to eight months per year against
573 three to four in the northwest, and very-high UV persists longest in the
574 south.

575 Critically, the categorical layer validated well against observed 2025 (Ta-
576 ble 6): exact-class agreement ranged from 81.5% (UV) to 89.8% (heat-index
577 comfort), and agreement to within one adjacent class reached 100% for all
578 five categorised conditions. The projection therefore reproduces not only the

Projected 2025-2026 annual-mean heat-stress conditions by district

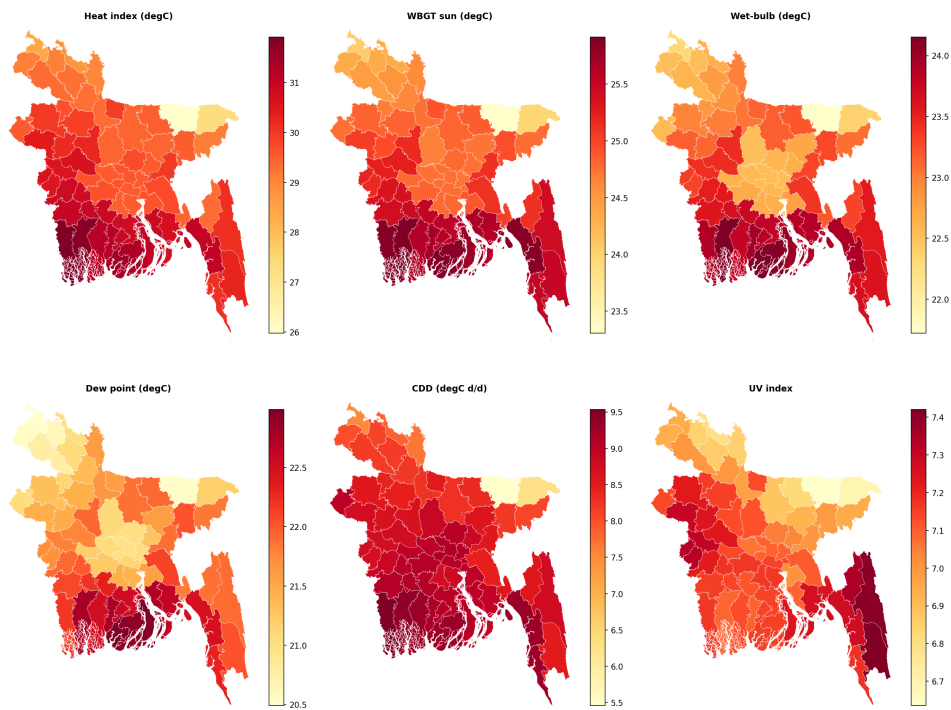


Figure 6: Projected 2025–2026 annual-mean heat-stress conditions by district: heat index, sun WBGT, wet-bulb temperature, dew point, cooling degree-days, and UV index. Every condition is shown at full district resolution rather than aggregated to zones.

continuous fields but also their operational risk classification.

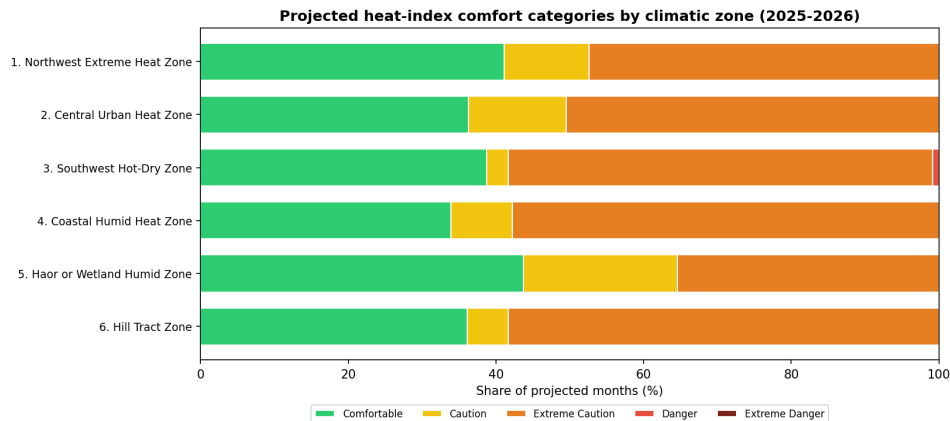


Figure 7: Projected heat-index comfort categories by climatic zone (share of projected months, 2025–2026). The Southwest Hot–Dry Zone carries the largest Extreme-Caution share and the only Danger occurrences; the Haor/Wetland Zone is mildest.

4.7. Projection uncertainty

The first-order error propagation yielded confidence bands that are approximately constant across the 24-month horizon, reflecting the dominance of the seasonal model’s error over the small trend extrapolation. For the Dhaka exemplar, the 95% heat-index band was $\pm 1.17^\circ\text{C}$, the wet-bulb band $\pm 0.99^\circ\text{C}$, and the sun-WBGT band $\pm 0.76^\circ\text{C}$. These bands exceed the projected inter-annual trend signal by more than an order of magnitude, and this relationship is stated explicitly to caution against over-interpreting small year-to-year differences in the projection.

Table 5: Projected risk-category distribution (share of projected district-months, 2025–2026, 64 districts). Only populated categories are shown.

Condition	Category	Share (%)
Heat-index comfort	Comfortable	37.8
	Caution	10.2
	Extreme Caution	51.8
	Danger	0.1
WBGT (sun, outdoor)	Low	59.0
	Moderate	37.0
	High	3.9
WBGT (shade, indoor)	Low	79.8
	Moderate	20.2
UV (WHO)	Moderate	4.2
	High	75.5
	Very High	20.4
Dew-point moisture stress	Comfortable	19.7
	Slightly Humid	6.6
	Humid	16.3
	Very Humid / Oppressive	11.9
	Extremely Oppressive	45.5

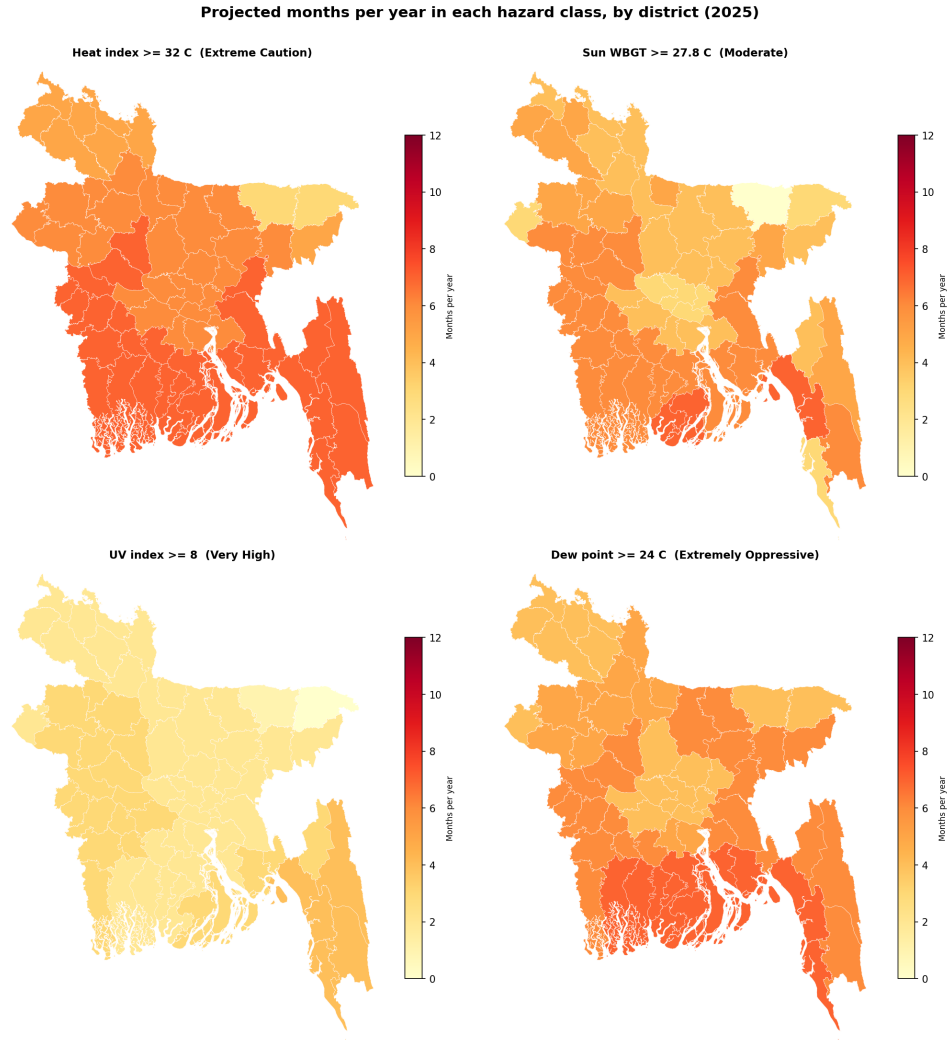


Figure 8: Projected number of months per year (of 12) that each district spends at or above a hazard threshold, for heat index ($\geq 32^{\circ}\text{C}$, Extreme Caution), sun WBGT ($\geq 27.8^{\circ}\text{C}$, Moderate), UV (≥ 8 , Very High), and dew point ($\geq 24^{\circ}\text{C}$, Extremely Oppressive). This months-per-year measure preserves the seasonal and spatial structure that a single dominant-class map collapses.

Table 6: Categorical validation against observed 2025: exact-class agreement and agreement to within one adjacent class, pooled over district-months.

Condition	n	Exact-class (%)	Within one class (%)
Heat-index comfort	704	89.8	100.0
WBGT (sun, outdoor)	704	84.1	100.0
WBGT (shade, indoor)	704	88.8	100.0
UV (WHO)	704	81.5	100.0
Dew-point moisture stress	704	87.1	100.0

5. Discussion

5.1. Principal findings

This study delivers district-resolved, uncertainty-aware near-term projections of seven temperature-correlated heat-stress conditions for all 64 districts of Bangladesh, together with a leakage-free external validation against independent 2025 observations. The recomputed thermal indices validated strongly heat index, wet-bulb temperature, and WBGT all at median $R^2 \geq 0.95$ and, when mapped onto their operational risk classes, reproduced the observed 2025 class in 81–90% of district-months exactly and in 100% to within one adjacent class. Because the indices are computed from the projected primary variables rather than predicted directly, this accuracy is achieved without target leakage and while preserving the physical coupling among temperature, humidity, and radiation. This validated, multi-index projection extends prior Bangladeshi heat-projection work that has largely treated single indices in isolation most notably the heat-index appraisal of Rahman et al. [20] and the WBGT trend analysis of Kamal et al. [9] and complements single-algorithm forecasting precedents such as Binte Ahmed

et al. [36] and Mohsin et al. [33] by covering all seven conditions jointly and validating them out-of-sample.

5.2. Skill must be interpreted correctly

A central and deliberately transparent result is that, although the ensembles comfortably outperform a seasonal-naive baseline, they are essentially at parity with monthly climatology (positive skill in only $\sim 32\%$ of district-variable cases). This is expected rather than disappointing: at monthly resolution the tropical seasonal cycle dominates the variance, and any competent learner reconstructs the climatological annual march that climatology also encodes. The contribution of the framework is therefore not a reduction in monthly error but the provision of a spatially complete, uncertainty-quantified, and trend-consistent projection including physiologically relevant indices and their operational risk classes that a bare climatology cannot supply. We state this explicitly to discourage over-claiming predictive skill that the data do not support. This transparency deliberately contrasts with the common practice of reporting high absolute accuracy for instance the R^2 values near 0.99 and 0.88–0.97 reported for Bangladeshi heat-index and temperature prediction by Binte Ahmed et al. [36] and Mohsin et al. [33] without benchmarking against a climatological reference, which can make routine seasonal reconstruction appear as genuine forecasting skill.

5.3. Physical and spatial interpretation

The observational analysis reveals a coherent but spatially structured warming signal. Maximum temperature increased in every district (median $+0.56^\circ\text{C}$ per decade) while minimum temperature remained essentially flat (median $+0.02^\circ\text{C}$ per decade), implying a widening diurnal temperature

631 range across much of the country a pattern with distinct implications for
 632 daytime heat exposure and nighttime physiological recovery. Humid-heat
 633 indices (heat index, wet-bulb, WBGT) rose in 57–58 of 64 districts, con-
 634 firming that the warming is not purely a dry-bulb phenomenon. Spatially,
 635 the Southwest Hot–Dry and Coastal Humid Heat zones carry the greatest
 636 projected burden across perceived heat, humid heat, and cooling-energy de-
 637 mand, whereas the elevated Haor/Wetland basin is consistently mildest; the
 638 coastal belt in particular combines the highest wet-bulb temperatures and
 639 cooling degree-days, marking it as a priority region. These patterns matter
 640 physiologically: the elevated coastal wet-bulb burden erodes the evaporative
 641 margin that Raymond et al. [6] identified as the hard limit of human heat
 642 tolerance, in the same coastal belt where Ekra et al. [21] found the steepest
 643 historical rise in heat discomfort. The widening diurnal range is also epi-
 644 demologically salient, since reduced nighttime cooling impairs physiological
 645 recovery and compounds the heatstroke-mortality and labour-capacity risks
 646 documented for Bangladesh and for outdoor workers more broadly [26, 57].

647 *5.4. Operational risk categories and their message*

648 Translating the projections into established risk classes sharpens the
 649 public-health narrative. The UV signal is the most uniformly hazardous con-
 650 dition: essentially the entire country falls in the WHO High class or above
 651 throughout the projection, and dew-point moisture stress reaches the Ex-
 652 tremely Oppressive class in almost half of all projected months. Heat-index
 653 comfort is dominated by the Extreme Caution class in the warm season, while
 654 the Danger class is essentially absent a direct and important consequence of
 655 categorising monthly means rather than daily maxima (Section 3.8), which
 656 the categories must be read against. The nationwide reach of the UV and hu-

657 midity hazards, contrasted with the more spatially variable heat-index and
658 WBGT burden, argues for differentiated adaptation responses: the near-
659 universal High-to-Very-High UV exposure calls for the sun-protection mea-
660 sures codified in the WHO UV-index guidance [17], whereas the humid-heat
661 burden is better addressed through the WBGT-based occupational-safety
662 framework of ISO 7243 [4, 14], whose relevance is amplified by the large
663 humid-heat labour losses estimated for South Asian outdoor workers [56].

664 *5.5. Methodological reflections*

665 Three design choices proved consequential. First, recomputing indices
666 from projected primary variables (rather than modelling them directly) both
667 prevents target leakage and guarantees internal physical consistency of the
668 projected fields. Second, the detrend–residual–retrend construction resolves
669 a failure mode of tree ensembles their inability to extrapolate a calendar-year
670 feature so that projected years evolve with the observed climate trend in-
671 stead of collapsing onto the last training year; imposing the robust 1980–2024
672 Sen slope additionally keeps the projection consistent with the manuscript’s
673 own trend analysis by construction. Third, the district choropleths ex-
674 posed a data-quality problem (early-record missing values encoded as zero)
675 that zonal-median summaries had masked, underscoring the value of full-
676 resolution spatial diagnostics and prompting the physical-floor correction
677 described in Section 3.12.

678 *5.6. Limitations*

679 Several limitations bound the interpretation. (i) The projection horizon
680 is deliberately restricted to 24 months; because the re-added trend is small
681 relative to the projection uncertainty (the 95% heat-index band exceeds the

682 annual trend signal by more than an order of magnitude), longer horizons
 683 would add deterministic drift without validatable information. (ii) Work-
 684 ing at monthly-mean resolution necessarily compresses daily extremes, so
 685 the risk categories describe typical monthly conditions and systematically
 686 under-represent short-duration dangerous events; a daily-resolution exten-
 687 sion is the natural next step. (iii) A composite heatstroke-hazard index was
 688 implemented but excluded from predictive results for negative out-of-sample
 689 skill, and is reported only descriptively. (iv) All meteorological inputs de-
 690 rive from a single commercial reanalysis-blended source; despite strong 2025
 691 validation, independent station-based corroboration would strengthen con-
 692 fidence, and the early-record inhomogeneity in a subset of districts (miti-
 693 gated here by masking) illustrates the sensitivity of long-run trends to input
 694 quality. (v) The models add little skill over climatology at monthly scale,
 695 as discussed above. The 24-month horizon is complementary to, rather
 696 than competitive with, the multi-decadal CMIP6 projections available for
 697 Bangladesh and the wider region [58, 44]: those studies bound the long-term
 698 trajectory under emission scenarios, whereas the present framework supplies
 699 the near-term, district-resolved, externally validated detail that dynamical
 700 downscaling cannot yet deliver at this resolution.

701 *5.7. Outlook*

702 Several extensions follow naturally. A daily-resolution implementation
 703 would recover the short-duration extremes that monthly means smooth away
 704 and let the risk categories reflect daily peaks the regime in which the heat-
 705 stroke indicator excluded here may regain skill [49]. Coupling the projected
 706 fields with health-surveillance and cooling-energy records would convert the
 707 physical outlook into quantified public-health and infrastructure burdens

[26]. Methodologically, the framework can absorb recent advances in the individual conditions reanalysis-based UV retrieval [51] and attention-based dew-point models with explainability [55] and its near-term projections can be systematically benchmarked against CMIP6 dynamical downscaling [58] to bridge near-term operational forecasting and long-term climate projection.

6. Conclusion

This study introduced the Temperature-Related Conditions (TRC) framework and applied it to project seven physiologically and operationally relevant heat-stress conditions across all 64 districts of Bangladesh. By training gradient-boosting ensembles on the 2014–2024 record with future-known seasonal features, re-imposing the observed 1980–2024 trend, and recomputing every thermal index from the projected primary variables, the framework produced district-level 24-month projections that validated strongly against independent 2025 observations median R^2 of 0.95–0.98 for the key indices, with operational risk classes reproduced to within one category in 100% of district-months.

The observational analysis revealed a coherent but spatially structured warming signal: maximum temperature rose in every district (median +0.56 °C per decade) while minimum temperature remained essentially flat, implying a widening diurnal temperature range, and the humid-heat indices increased across most of the country. Spatially, the coastal and southwestern zones emerged as the areas of greatest combined perceived-heat, humid-heat, and cooling-energy burden, whereas ultraviolet and oppressive-humidity hazards were effectively nationwide. Crucially, the ensembles were found to be at parity with monthly climatology; the framework’s contribution is therefore

not improved monthly accuracy but a spatially complete, uncertainty-aware, and trend-consistent outlook delivered at full district resolution that supports targeted heat-adaptation planning. One indicator, heatstroke risk, was excluded from the predictive results for a lack of out-of-sample skill and reported only descriptively.

Several directions follow naturally. Extending the analysis to daily resolution would capture the short-duration extremes that monthly means smooth away and would allow the risk categories to reflect daily peaks rather than typical monthly conditions. Coupling the projected indices with health-surveillance and energy-demand records would translate this physical outlook into quantified public-health and infrastructure burdens. More broadly, the leakage-free, trend-consistent, and externally validated design demonstrated here offers a transferable template for district-level heat-stress projection in other data-constrained, climatically heterogeneous regions.

CRedit authorship contribution statement

Mohammad Khalid Hasan Nabil: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Project administration.

Tahia Parsha: Investigation, Software, Writing – original draft, Writing – review & editing.

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756 Declaration of competing interest

757 The authors declare that they have no known competing financial inter-
758 ests or personal relationships that could have appeared to influence the work
759 reported in this paper.

760 Data availability

761 The raw daily meteorological data are available from the Visual Cross-
762 ing Weather API (<https://www.visualcrossing.com>). The complete de-
763 terministic machine-learning pipeline source code, the interactive Stream-
764 lit web-application dashboard, and the comprehensive 64-district projected
765 dataset are permanently archived on Zenodo ([https://doi.org/10.5281/](https://doi.org/10.5281/zenodo.21418291)
766 [zenodo.21418291](https://doi.org/10.5281/zenodo.21418291)). The interactive dashboard which visualises the projec-
767 tions, validation, and risk categories for all 64 districts and seven conditions is
768 served live at [https://tcc-64-districts-mohammad-khalid.streamlit.](https://tcc-64-districts-mohammad-khalid.streamlit.app/)
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